

A Short Primer on Vector Databases

What is a vector database? A vector database is a specialized data store that indexes high-dimensional embedding vectors to support similarity search. Instead of exact-match queries, users ask for the k nearest vectors to a query vector under a distance metric such as cosine or L2.

Indexing strategies. Modern vector stores use approximate-nearest-neighbor indexes such as HNSW (Hierarchical Navigable Small World) or IVF-PQ (Inverted File with Product Quantization). HNSW trades memory for fast low-latency lookups; IVF-PQ trades recall for a much smaller memory footprint and is preferred for billion-scale collections.

Use cases. The most common use case is retrieval-augmented generation (RAG), where relevant document chunks are retrieved by vector similarity and injected into a language model prompt. Other uses include recommendation systems, deduplication, and semantic image search.

Choosing a store. For local development, Chroma and FAISS are the most popular options because they run in-process with no network dependency. For production, managed services like Pinecone, Weaviate Cloud, and pgvector on PostgreSQL are common. The query API is broadly similar across all of them — the main differences are operational.